Experiment:6

Spectrum and Bandwidth Analysis of a Given Signal

Task B - Simplified AM signal • Use carrier 100 Hz, modulating frequency 10 Hz and modulation index 0.5. • Identify carrier, sidebands and transmission bandwidth.

```
clc;
clear;
close all;
%
==========================================================
=
% Simplified AM Signal - Spectrum and Bandwidth Analysis
%
==========================================================
=
% Parameters
Ac = 1;             % Carrier amplitude
fc = 100;           % Carrier frequency (Hz)
fm = 10;            % Modulating frequency (Hz)
mu = 0.5;           % Modulation index
% Sampling frequency
Fs = 5000;          % Sampling frequency (Hz)
% Time vector
t = 0:1/Fs:1;
```

```
% Message signal
m = cos(2*pi*fm*t);
% Carrier signal
c = Ac*cos(2*pi*fc*t);
% AM signal
s = Ac*(1 + mu*m).*cos(2*pi*fc*t);
%
==========================================================
=
% Display parameters
%
==========================================================
=
fprintf('=============================================\n');
fprintf('        SIMPLIFIED AM SIGNAL ANALYSIS\n');
fprintf('=============================================\n');

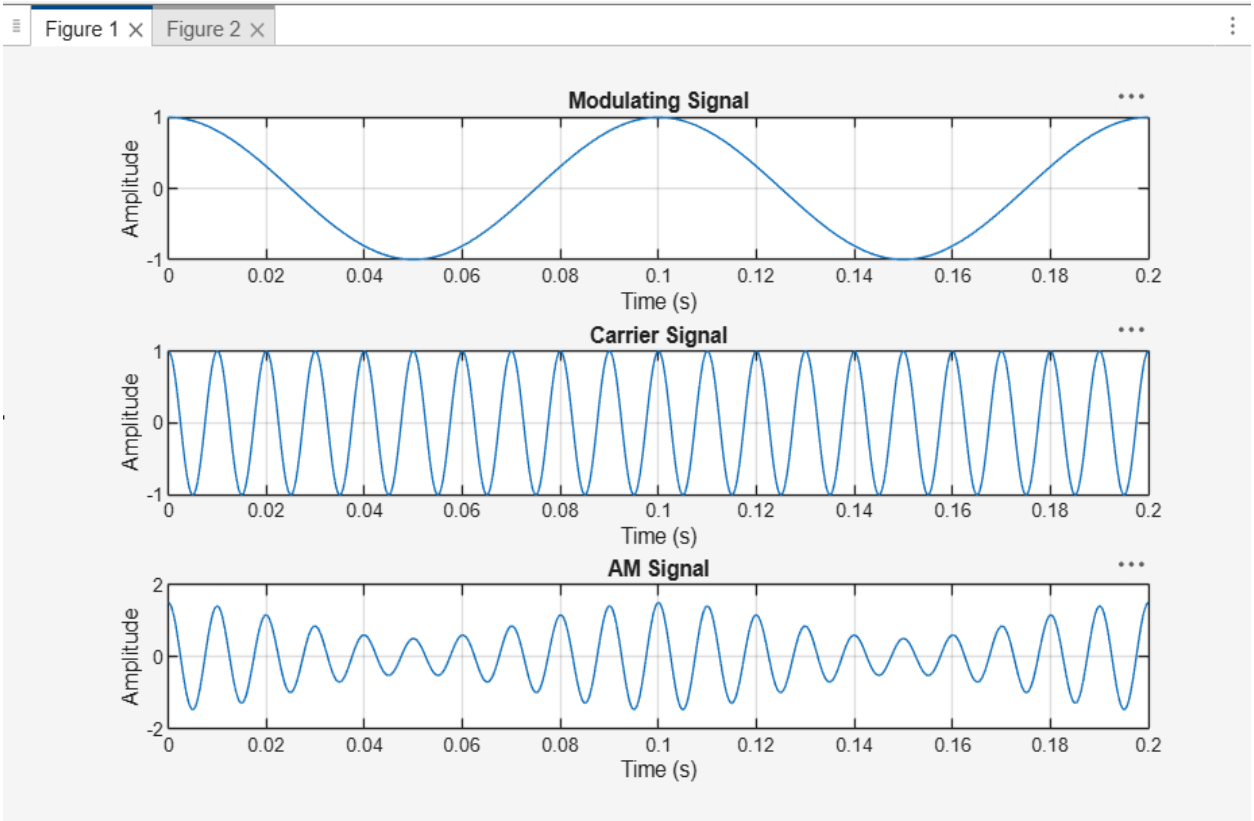
fprintf('Carrier Frequency      = %d Hz\n', fc);
fprintf('Modulating Frequency   = %d Hz\n', fm);
fprintf('Modulation Index       = %.2f\n', mu);

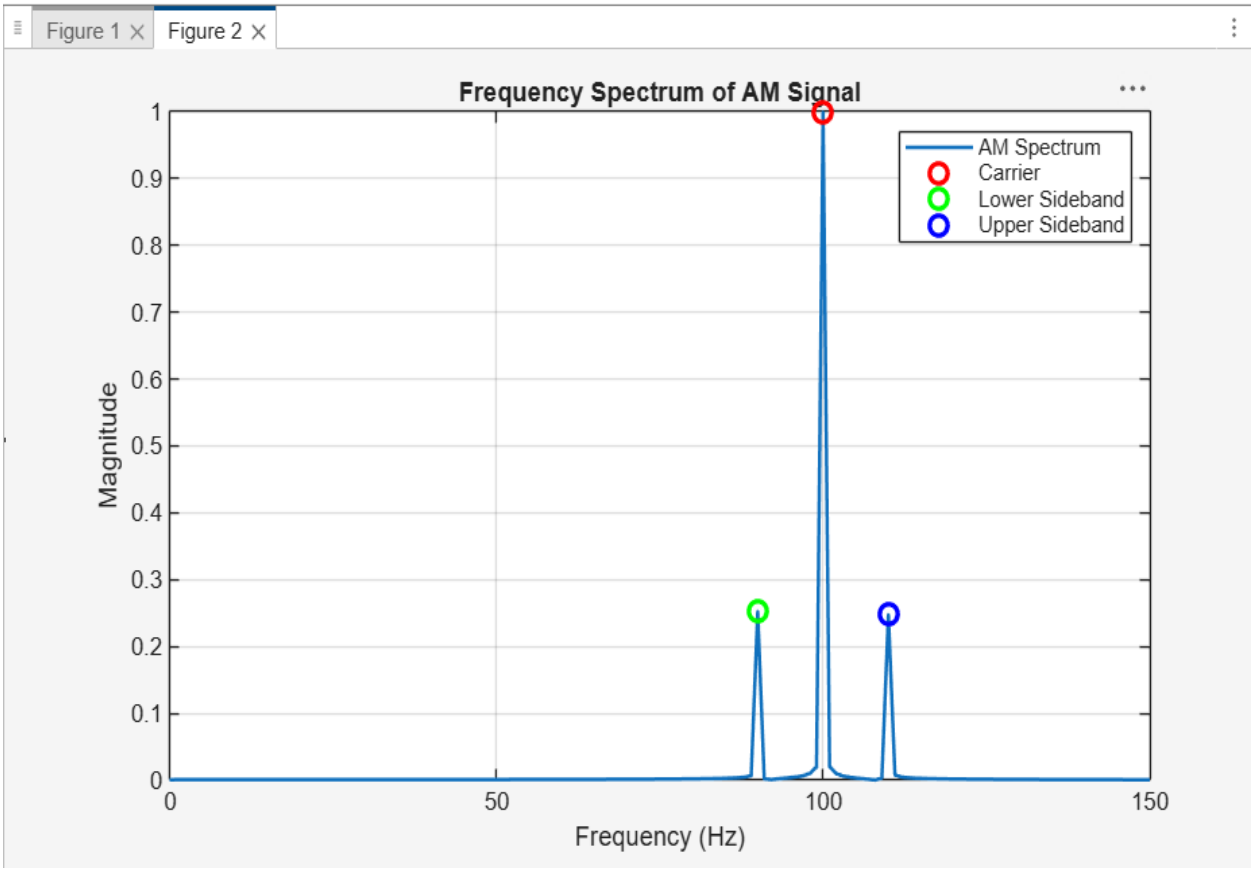
% Sideband frequencies
LSB = fc - fm;
USB = fc + fm;
% Transmission bandwidth
```

```
BW = USB - LSB;
fprintf('Lower Sideband (LSB)   = %d Hz\n', LSB);
fprintf('Carrier Frequency      = %d Hz\n', fc);
fprintf('Upper Sideband (USB)   = %d Hz\n', USB);
fprintf('Transmission Bandwidth  = %d Hz\n', BW);
fprintf('================================================\n');
%
==========================================================
=
% Time-domain plots
%
==========================================================
=
figure;
subplot(3,1,1);
plot(t, m);
xlim([0 0.2]);
grid on;
title('Modulating Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,2);
plot(t, c);
xlim([0 0.2]);
grid on;
```

```
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(t, s);
xlim([0 0.2]);
grid on;
title('AM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
% Frequency Spectrum using FFT
N = length(s);
% FFT
S = fft(s);
% Two-sided spectrum
P2 = abs(S/N);
% Single-sided spectrum
P1 = P2(1:N/2+1);
P1(2:end-1) = 2*P1(2:end-1);
% Frequency axis
f = Fs*(0:(N/2))/N;
%
==========================================================
=
% Plot AM Spectrum
```

```
%
==========================================================================
=
figure;
plot(f, P1, 'LineWidth', 1.5);
xlim([0 150]);
grid on;
title('Frequency Spectrum of AM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
% Mark Carrier and Sidebands
hold on;
plot(fc, P1(find(abs(f-fc)==min(abs(f-fc)))), 'ro', ...
    'MarkerSize', 8, 'LineWidth', 2);
plot(LSB, P1(find(abs(f-LSB)==min(abs(f-LSB)))), 'go', ...
    'MarkerSize', 8, 'LineWidth', 2);
plot(USB, P1(find(abs(f-USB)==min(abs(f-USB)))), 'bo', ...
    'MarkerSize', 8, 'LineWidth', 2);
legend('AM Spectrum', 'Carrier', 'Lower Sideband', ...
       'Upper Sideband');
hold off;
```


Figure 1
Figure 2
Modulating Signal
Amplitude
Time (s)
Carrier Signal
Amplitude
Time (s)
AM Signal
Amplitude
Time (s)


Figure 1
Figure 2
Frequency Spectrum of AM Signal
AM Spectrum
Carrier
Lower Sideband
Upper Sideband
Magnitude
Frequency (Hz)

Experiment :7

Pole-Zero and Stability Analysis of Continuous Time Systems

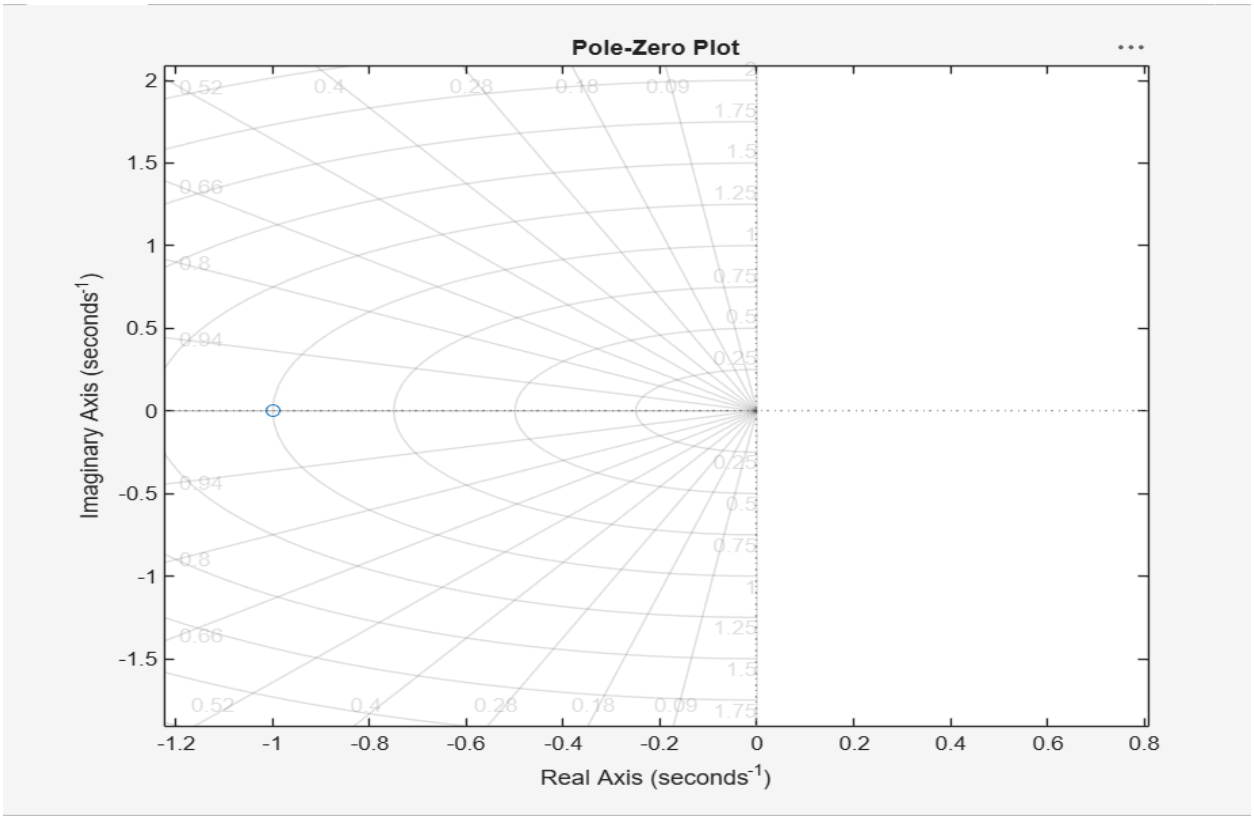
Task C - Unstable oscillatory system • H(s)=(s+1)/(s^2-2s+5). • Explain why positive real parts produce growth under the causal assumption.

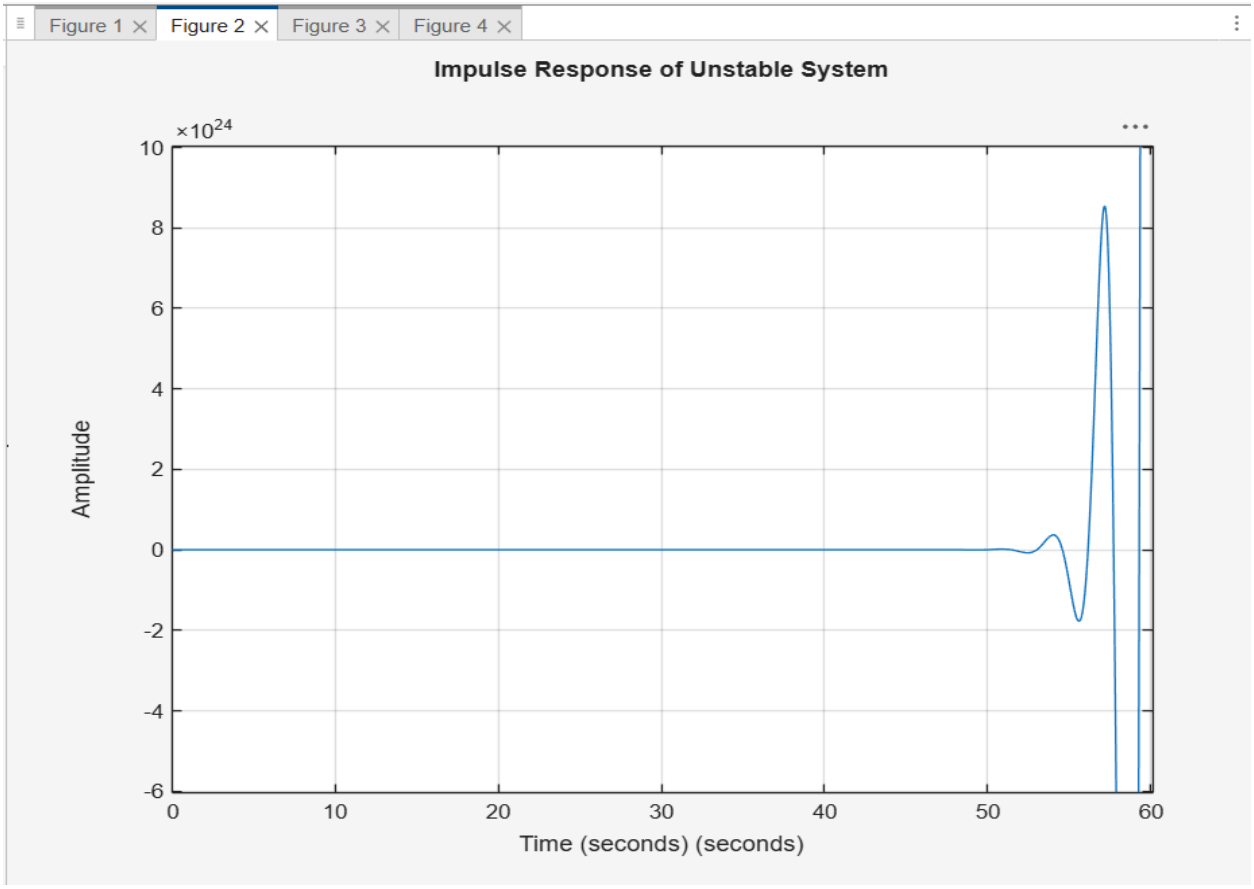
```
clc;
clear;
close all;
%
==========================================================
=
% TASK C - UNSTABLE OSCILLATORY SYSTEM
% H(s) = (s+1)/(s^2 - 2s + 5)
%
==========================================================
=
% Define numerator and denominator
num = [1 1];
den = [1 -2 5];
% Create transfer function
H = tf(num, den);
% Display transfer function
disp('Transfer Function:');
H
% Find poles
```

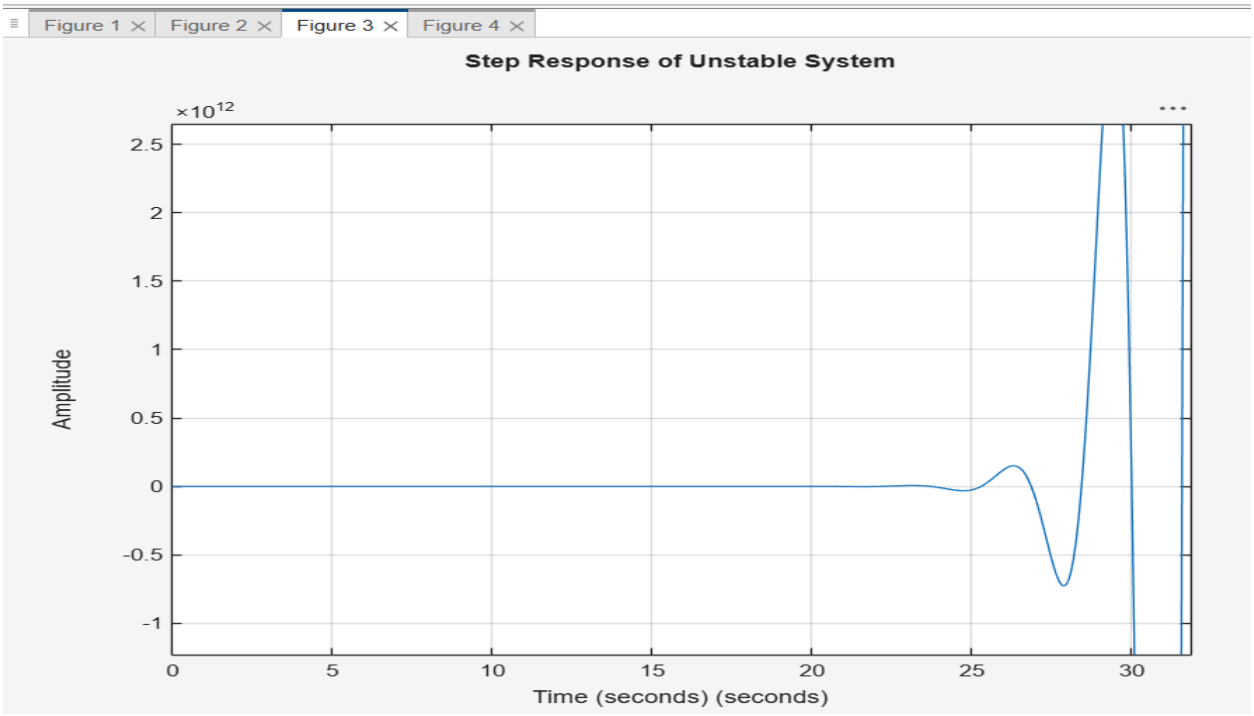
```
p = pole(H);
disp('Poles of the system:');
disp(p);
% Display poles
fprintf('\nPole 1 = %.2f + j%.2f\n', real(p(1)), imag(p(1)));
fprintf('Pole 2 = %.2f + j%.2f\n', real(p(2)), imag(p(2)));
%
==========================================================
=
% 1. POLE-ZERO PLOT
%
==========================================================
=
figure;
pzmap(H);
grid on;
title('Pole-Zero Plot');
xlabel('Real Axis');
ylabel('Imaginary Axis');
%
==========================================================
=
% 2. IMPULSE RESPONSE
%
==========================================================
=
figure;
```

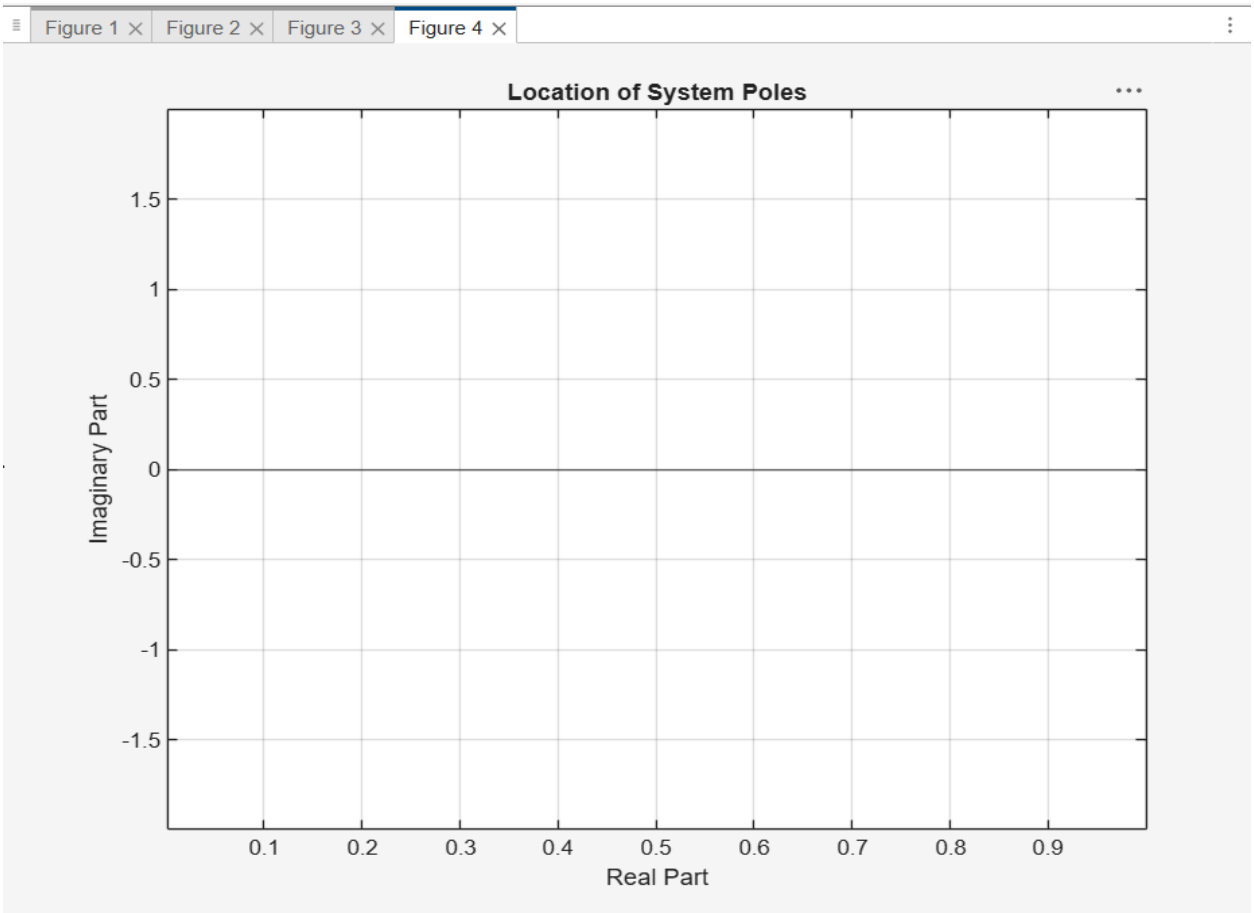
```
impulse(H);
grid on;
title('Impulse Response of Unstable System');
xlabel('Time (seconds)');
ylabel('Amplitude')
%
==========================================================
=
% 3. STEP RESPONSE
%
==========================================================
figure;
step(H);
grid on;
title('Step Response of Unstable System');
xlabel('Time (seconds)');
ylabel('Amplitude');
%
==========================================================
=
% 4. POLE LOCATION USING COMPLEX PLANE
%
==========================================================
=
figure;
plot(real(p), imag(p), 'x', 'MarkerSize', 12, 'LineWidth', 2);
```

```
grid on;
hold on;
% Draw real and imaginary axes
xline(0);
yline(0);
title('Location of System Poles');
xlabel('Real Part');
ylabel('Imaginary Part');
hold off;
%
==========================================================================
=
% STABILITY CHECK
%
==========================================================================
=
if any(real(p) > 0)
    fprintf('\nSystem is UNSTABLE.\n');
    fprintf('The poles have positive real parts.\n');
    fprintf('Therefore, the response grows exponentially with time.\n');
end
```


Pole-Zero Plot
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
2
1.5
1
0.5
0
-0.5
-1
-1.5
-1.2
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8


Figure 1
Figure 2
Figure 3
Figure 4
Impulse Response of Unstable System
×10^{24}
Amplitude
Time (seconds) (seconds)
10
8
6
4
2
0
-2
-4
-6
0
10
20
30
40
50
60


Figure 1
Figure 2
Figure 3
Figure 4
Step Response of Unstable System
×10^12
2.5
2
1.5
1
0.5
0
-0.5
-1
Amplitude
0
5
10
15
20
25
30
Time (seconds) (seconds)


Figure 1
Figure 2
Figure 3
Figure 4
Location of System Poles
1.5
1
0.5
0
-0.5
-1
-1.5
Imaginary Part
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Real Part

```
clc;
clear;
close all;
% Time vector
t = -2:0.001:4;
% Unit step function
u = double(t>=0);
% Original signal x(t)=exp(-t)u(t)
x = exp(-t).*u;
% Transformed signal y(t)=2x(2t-1)
t1 = 2*t - 1;
u1 = double(t1>=0);
y = 2*exp(-(t1)).*u1;
% Plotting
figure;
subplot(2,1,1)
plot(t,x,'b','LineWidth',2)
grid on
xlabel('Time (t)')
ylabel('x(t)')
title('Original Signal  x(t)=e^{-t}u(t)')
```

```
subplot(2,1,2)
plot(t,y,'r','LineWidth',2)
grid on
xlabel('Time (t)')
ylabel('y(t)')
title('Transformed Signal  y(t)=2x(2t-1)')
```

